

The Texas Instruments (TI) company manufactures the most popular scientific calculators used by electrical engineering students today and has been doing so for many years. With the advent of the TI-85 in the mid 1990s and its built-in complex number calculations capability, TI scientific calculators have caused dramatic changes in electrical engineering courses that require complex number calculations, particularly courses on AC circuits, AC power systems, and AC electric machinery. The TI scientific calculator allows students to more easily, quickly, and accurately perform the detailed complex number calculations required in these courses, and is now considered an essential tool in these courses.

This document is intended to provide an overview of typical calculator tasks required in electrical engineering courses. Today, most electrical engineering students own the TI-89 calculator or a variant of the TI-89. So the syntax in this document will be specific to the TI-89 calculator. However this document can be adapted easily to any other calculator that has the same functionality.

1. Optimal Setting of Calculator Modes

The first thing that should be done with a scientific calculator used for electrical engineering courses is to customize its mode settings. To access the mode settings menu in the TI-89 calculator, simply press the **[MODE]** key. You will see many options in this menu that can be changed. You should make the following settings changes to put your calculator in the best possible mode for electrical engineering course work.

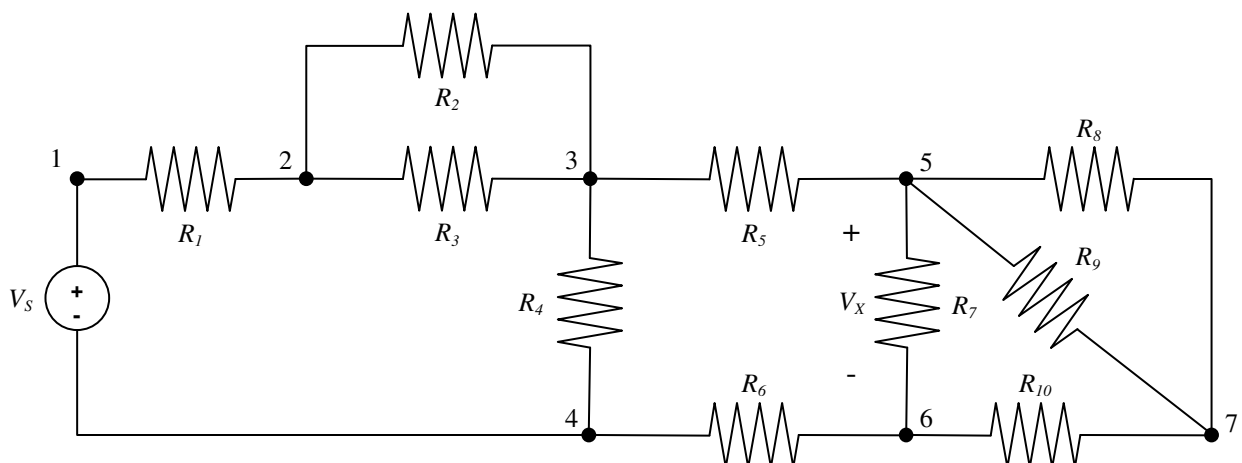
- The **Angle** option should be set to **DEGREE**.
- The **Complex Format** option should be set to **POLAR**.
- The **Exact/Approx** option should be set to **APPROXIMATE**.

2. Storing Numeric Constants and Intermediate Calculation Results as Variables

A critical timesaving and accuracy practice is to store frequently used numeric constants and intermediate calculation results as variables in the calculator. Storing numeric constants that are frequently used completely eliminates the chance of making a mistake entering the numeric constant during a test and saves valuable time. Storing intermediate calculation results eliminates the time consuming and error prone process of having to manually write down intermediate calculation results with many decimal places and then having to key in these written down values again later in the problem when they are needed.

Developing a habit of storing intermediate calculation results as variables in your calculator can greatly improve your problem solving efficiency and accuracy. This is particularly important when doing complex number calculations that require twice the amount of writing and twice the amount of button pressing on your calculator to enter.

A good example to illustrate the benefits of storing intermediate calculations results is in the analysis of a single source series/parallel circuit. Consider the simple DC circuit below for which you are asked to calculate the output voltage V_x by first reducing the circuit back to the source and then using the calculated source current to work back through the circuit to calculate the desired output voltage.



The numeric values for the voltage source and the resistors in the circuit are given below.

$$V_s = 154.25 \text{ V}, \quad R_1 = 5.67 \text{ k}\Omega, \quad R_2 = 25.94 \text{ k}\Omega, \quad R_3 = 31.81 \text{ k}\Omega, \quad R_4 = 25.62 \text{ k}\Omega$$

$$R_5 = 595 \text{ }\Omega, \quad R_6 = 3.78 \text{ }\Omega, \quad R_7 = 21.25 \text{ k}\Omega, \quad R_8 = 15.56 \text{ k}\Omega, \quad R_9 = 18.11 \text{ k}\Omega, \quad R_{10} = 2.25 \text{ k}\Omega$$

To calculate the desired output voltage, the circuit must first be reduced from left to right to a point that will allow calculating the source current. This is done using the equations below.

$$R_{56} = \left\{ R_7^{-1} + \left[\left(R_8^{-1} + R_9^{-1} \right)^{-1} + R_{10} \right]^{-1} \right\}^{-1} = \text{_____} \Omega$$

$$R_{34} = \left[R_4^{-1} + \left(R_5 + R_{56} + R_6 \right)^{-1} \right]^{-1} = \text{_____} \Omega$$

$$R_{23} = \left(R_2^{-1} + R_3^{-1} \right)^{-1} = \text{_____} \Omega$$

Next using the reduced circuit variables above, the source current is calculated and then the desired output voltage V_x is calculated. This is done using the equations below.

$$I_s = V_s \left(R_1 + R_{23} + R_{34} \right)^{-1} = \text{_____} \text{ A}$$

$$V_{34} = R_{34} I_s = \text{_____} \text{ V}$$

$$V_x = V_{34} R_{56} / \left(R_5 + R_{56} + R_6 \right) = \text{_____} \text{ V}$$

As an exercise, use a clock or watch to measure the time it takes you to plug the numbers into the above equations and write down the results in the blank spaces above to the right of the equations. When you do this, **write down the intermediate results rounded off to only three decimal places, and use these written down three decimal place values to calculate other quantities that need the values.** Record your elapsed time to completion in the space below.

Elapsed Time = _____ seconds

Both the time required to complete the work above and the accuracy of the results obtained can be improved by storing intermediate calculation results as variables in your calculator instead of writing down and then re-entering truncated values. To store an entered number or a calculation result as a variable, simply enter the number or perform the calculation, and then press the **[STO▶]** key. When the **[STO▶]** key is pressed, the calculator automatically goes into alphabetic mode and is waiting for you to enter an alphanumeric variable name. After typing the desired variable name, pressing the **[ENTER]** key will store the value in the calculator's memory with the entered variable name.

Variable names must start with an alphabetic character but can also contain numbers, for example XYZ or R123 or Z5A1 are acceptable variable names. To switch back and forth from alphabetic mode to numeric mode when entering a variable name, use the **[alpha]** key. As an exercise, perform the calculation 4 times pi times 1×10^{-7} and then store the result as the variable name M0 in your calculator.

In the TI-89 and TI-92 calculators, there are several reserved variables names that you cannot use for storing numbers, and unfortunately some of these reserved variable names are ones we like to use in electrical engineering problems. These reserved variable names that you cannot use are R, R1 through R99, ZC, Z1 through Z99, YC, and Y1 through Y99. You can avoid using these exact reserved variable names by simply adding another character to the variable name. For example use RR1 instead of R1, or use ZZ1 instead of Z1, or use YCC instead of YC. This variable name constraint only exists in the TI-89 and TI-92 calculators. Interestingly, the older TI-85 and TI-86 calculators do not have such a constraint, and they allow you to use the specific variable names mentioned above that you cannot use on the TI-89 and TI-92 calculators.

Next to demonstrate the benefits of storing intermediate calculations results, repeat the previous example but this time using your calculator to store intermediate values as variables and then recalling these variables when needed instead of keying in the intermediate calculation results.

Just like the first time the exercise was done, use a clock or watch to measure the time it takes you to plug the numbers into the equations and to write down the results in the blank spaces to the right of the equations. Again you should **write down the results with only three decimal places**. However this time **also store the results of each equation as a unique variable in your calculator**. The variable names you enter into your calculator should **match the mathematical variables in the equations as closely as possible**. When previously calculated intermediate values are required in subsequent equations, **do not manually key in the written down numeric value**. Instead use the alpha key and **enter the variable name** for which the intermediate value was stored.

The numeric values for the voltage source and the resistors in the circuit are given below again.

$$V_s = 154.25 \text{ V}, \quad R_1 = 5.67 \text{ k}\Omega, \quad R_2 = 25.94 \text{ k}\Omega, \quad R_3 = 31.81 \text{ k}\Omega, \quad R_4 = 25.62 \text{ k}\Omega$$

$$R_5 = 595 \text{ }\Omega, \quad R_6 = 3.78 \text{ }\Omega, \quad R_7 = 21.25 \text{ k}\Omega, \quad R_8 = 15.56 \text{ k}\Omega, \quad R_9 = 18.11 \text{ k}\Omega, \quad R_{10} = 2.25 \text{ k}\Omega$$

The circuit analysis equations are given below again.

$$R_{56} = \left\{ R_7^{-1} + \left[\left(R_8^{-1} + R_9^{-1} \right)^{-1} + R_{10} \right]^{-1} \right\}^{-1} = \text{_____} \Omega$$

$$R_{34} = \left[R_4^{-1} + \left(R_5 + R_{56} + R_6 \right)^{-1} \right]^{-1} = \text{_____} \Omega$$

$$R_{23} = \left(R_2^{-1} + R_3^{-1} \right)^{-1} = \text{_____} \Omega$$

$$I_s = V_s \left(R_1 + R_{23} + R_{34} \right)^{-1} = \text{_____} \text{ A}$$

$$V_{34} = R_{34} I_s = \text{_____} \text{ V}$$

$$V_x = V_{34} R_{56} / \left(R_5 + R_{56} + R_6 \right) = \text{_____} \text{ V}$$

$$\text{Elapsed Time} = \text{_____} \text{ seconds}$$

Now compare the elapsed time just recorded with your previously recorded elapsed time. You should see an improvement. More importantly though, compare the accuracy of the final result for the calculated output voltage V_x . Since intermediate values were not manually truncated to three decimal places the second time the calculations were done, the final result obtained is far more accurate. If you followed the instructions correctly and performed the calculations correctly, you should have gotten a final result of 31.207V the first time and then 31.340V the second time.

3. Entering Complex Numbers

In the electrical engineering areas of AC circuits, power system analysis, and electric machinery, numeric quantities worked with are complex numbers and complex number calculations are required to perform analysis. A good scientific calculator is an amazing and indispensable tool in these areas.

Complex numbers can be entered into your calculator in either rectangular form ($a+jb$) or polar form ($r\angle\theta$). For either form, surrounding parentheses tell the calculator to recognize it as a complex number.

To enter the rectangular form complex number $2-j5$, use the following sequence of keystrokes:

(2 - 2nd CATALOG 5)

To enter the rectangular form complex number $-10+j7$, use the following sequence of keystrokes:

$\boxed{(-)}\boxed{1}\boxed{0}\boxed{+}\boxed{2nd}\boxed{CATALOG}\boxed{7}\boxed{)}$

To enter the polar form complex number $120\angle-12^\circ$, use the following sequence of keystrokes:

$\boxed{1}\boxed{2}\boxed{0}\boxed{2nd}\boxed{EE}\boxed{(-)}\boxed{1}\boxed{2}\boxed{)}$

As an exercise, first enter the following 5 constants and store them as variables in your calculator using the exact same variable names (V1, ZL, A, ZX1, ZX2) in your calculator as used in the equations below.

$$\bar{V}_1 = 2298\angle 0^\circ, \quad \bar{Z}_L = 1.96\angle 69.62^\circ, \quad a = \frac{2200}{440}, \quad \bar{\mathcal{J}}_{x1} = 0.09 + j0.71, \quad \bar{\mathcal{J}}_{x2} = 0.01 + j0.11$$

Next enter the following 5 intermediate calculation equations into your calculator that use the stored values, and store each equation result as a variable in your calculator using the exact same variable names (ZX, Z1EQ, I1, I2, V2) in your calculator as used in the equations below.

$$\bar{Z}_x = \bar{\mathcal{J}}_{x1} + a^2 \bar{\mathcal{J}}_{x2}, \quad \bar{Z}_{1eq} = \bar{Z}_x + a^2 \bar{Z}_L, \quad \bar{I}_1 = \bar{V}_1 / \bar{Z}_{1eq}, \quad \bar{I}_2 = a \bar{I}_1, \quad \bar{V}_2 = \bar{Z}_L \bar{I}_2$$

Next recall the stored calculation results one at a time by typing their stored variable names (ZX, Z1EQ, I1, I2, V2) and pressing the $\boxed{ENTER}$ key. Your results should match the values displayed below.

$$\begin{aligned} \bar{Z}_x &= 3.4767\angle 84.388^\circ, & \bar{Z}_{1eq} &= 52.36932\angle 70.590^\circ, & \bar{I}_1 &= 43.88065\angle -70.590^\circ \\ \bar{I}_2 &= 219.4033\angle -70.590^\circ, & \bar{V}_2 &= 430.0304\angle -0.970^\circ \end{aligned}$$

4. Complex Variable Functions

The TI calculator has several built-in complex variable functions that are accessible in the **MATH** menu and are accessed by pressing $\boxed{2nd}\boxed{5}$ on the TI-89 and TI-92 calculators. There is a **COMPLEX** submenu in the **MATH** menu that contains the complex variable functions that must then be selected.

The complex variable functions available on the calculator include:

- Real part of a complex number
- Imaginary part of a complex number
- Magnitude of a complex number
- Phase angle of a complex number
- Conjugate of a complex number
- Convert to rectangular
- Convert to polar.

These complex variable functions are all extremely useful in the areas of power system analysis and electric machinery.

As an exercise, use the stored variables from the previous example to perform the following additional calculations that use your calculator's built-in complex variable functions.

$$\begin{aligned} \bar{S}_2 &= \bar{V}_2 \cdot \bar{I}_2^* && \text{evaluate using the complex conjugate function and store as calculator variable S2} \\ P_2 &= \text{Re}\{\bar{S}_2\} && \text{evaluate using the real part function and store as calculator variable P2} \\ Q_2 &= \text{Im}\{\bar{S}_2\} && \text{evaluate using the imaginary part function and store as calculator variable Q2} \\ pf_2 &= \cos(\angle \bar{S}_2) && \text{evaluate using the angle function and store as calculator variable PF2} \end{aligned}$$

Complex Variable Hyperbolic Functions

In certain transmission line modeling calculations, hyperbolic functions with complex number arguments must be evaluated. The TI-89/92 has an internal software bug that produces a domain error when doing these calculations, unless they are done in a very specific way. This very specific way entails using complex number arguments that have their angle in radians instead of degrees. The following 3-step process explains exactly how to do this on the TI-89/92 calculator.

Step 1:

Change the calculator complex number mode from degrees to radians. Then enter the complex number, but append a ($^\circ$) to the angle using the keystrokes $\boxed{2\text{nd}}\boxed{5}\boxed{1}\boxed{2}$.

Step 2:

Compute the $\sinh()$ or $\cosh()$ of this complex number using the MATH menu. The result will be displayed with an angle in radians. Extract the magnitude of the result and store it as a variable. Extract just the angle from the previous result using the keystrokes $\boxed{2\text{nd}}\boxed{5}\boxed{5}\boxed{4}$ and store as a variable.

Step 3:

Change the calculator complex number mode from radians back to degrees. Build the result in the following form $(A\angle B^\circ)$, where A is the stored magnitude and B is the stored angle. Since the angle is in radians you will need to append the 'r' which can be found by $\boxed{2\text{nd}}\boxed{5}\boxed{2}\boxed{2}$. The final displayed result will be a complex number with an angle in degrees.

The above 3-step procedure can be automated on the TI-89/92 by creating a simple program and then using this program whenever it's needed. To enter such a program, go to $\boxed{\text{APPS}}$, then [Program Editor], and then [New...]. Then select [Program], and enter the program code listed below.

```
hypxz(yy)
Prgm
abs(yy) → m
angle(yy) → a
setMode("Angle","RADIAN")*
(m ∠ a°) → yy
sinh(yy) → zz
cosh(yy) → xx
abs(zz) → m
abs(xx) → m1
angle(zz) → a
angle(xx) → a1
setMode("Angle","DEGREE")**
(m ∠ ar) → zz
(m1 ∠ a1r) → xx
EndPrgm
```

Notes:

* $\boxed{\text{F6}}\boxed{3}\boxed{1}\boxed{\text{ENTER}}\boxed{\text{ENTER}}$
 ** $\boxed{\text{F6}}\boxed{3}\boxed{2}\boxed{\text{ENTER}}\boxed{\text{ENTER}}$

Additional Notes:

$^\circ$ This character can be found by pressing: $\boxed{2\text{nd}}\boxed{1}$
^r This character can be found by pressing: $\boxed{2\text{nd}}\boxed{5}\boxed{2}\boxed{2}$

The program name is hypxz. Before running the program, the complex number you want to calculate the hyperbolic functions of must be entered and stored in variable yy. The results of the program are stored in variables xx and zz, where xx contains the calculated cosh(yy) and zz contains the calculated sinh(yy). **Be aware that if you previously had numbers stored in variables xx or zz, they will be overwritten by this program without warning!**

An example of calculating hyperbolic functions is given below, in which the ABCD parameters for a special type of transmission line model, called the long length model, are calculated.

A transmission line has the following impedance characteristics:

$$R = 0.03366 \, \Omega / mi, \quad L = 1.412038 \, mH / mi, \quad G = 0 \, \text{S} / mi, \quad C = 20.773266 \, nF / mi, \quad \ell = 75 \, mi$$

First the following distributed impedance parameters are calculated:

$$\begin{aligned}\bar{\mathcal{Z}} &= R + j\omega L = 0.533389 \angle 86.381^\circ \, \Omega / mi, & \bar{Y} &= G + j\omega C = 7.831337 \times 10^{-6} \angle 90^\circ \, \Omega^{-1} / mi \\ \bar{\gamma} &= \sqrt{\bar{\mathcal{Z}} \bar{Y}} = 2.043808 \times 10^{-3} \angle 88.191^\circ / mi, & \bar{Z}_c &= \sqrt{\bar{\mathcal{Z}} / \bar{Y}} = 260.978160 \angle -1.809^\circ \, \Omega\end{aligned}$$

Next if your calculator can handle computing hyperbolic functions with complex arguments, then:

$$\begin{aligned}\bar{A} = \bar{D} &= \cosh(\bar{\gamma}\ell) = 0.988298 \angle 0.043^\circ & \bar{B} &= \bar{Z}_c \sinh(\bar{\gamma}\ell) = 39.84802743 \angle 86.395^\circ \\ \bar{C} &= [\sinh(\bar{\gamma}\ell)] / \bar{Z}_c = 5.850574 \times 10^{-4} \angle 90.014^\circ\end{aligned}$$

Else the hyperbolic functions can be computed in the alternate way shown below:

$$\bar{\gamma}\ell = (2.043808 \times 10^{-3} \angle 88.191^\circ / mi)(75 \, mi) = 0.1532856 \angle 88.191^\circ$$

$$\bar{\gamma}\ell = \alpha\ell + j\beta\ell = 0.004839 + j0.153209$$

$$\alpha\ell = 0.004839, \quad \beta\ell = 0.153209 \, rad = 8.778^\circ$$

$$\begin{aligned}\sinh(\bar{\gamma}\ell) &= \sinh(\alpha\ell) \cos(\beta\ell) + j \cosh(\alpha\ell) \sin(\beta\ell) \\ &= (0.004839)(0.988287) + j(1.000012)(0.152606) \\ &= 0.004782 + j0.152608 = 0.152683 \angle 88.208^\circ\end{aligned}$$

$$\begin{aligned}\cosh(\bar{\gamma}\ell) &= \cosh(\alpha\ell) \cos(\beta\ell) + j \sinh(\alpha\ell) \sin(\beta\ell) \\ &= (1.000012)(0.988287) + j(0.004839)(0.152606) \\ &= 0.988299 + j0.000738 = 0.988298 \angle 0.043^\circ\end{aligned}$$

$$\bar{A} = \bar{D} = \cosh(\bar{\gamma}\ell) = 0.988298 \angle 0.043^\circ \quad \bar{B} = \bar{Z}_c \sinh(\bar{\gamma}\ell) = 39.84802743 \angle 86.395^\circ$$

$$\bar{C} = [\sinh(\bar{\gamma}\ell)] / \bar{Z}_c = 5.850574 \times 10^{-4} \angle 90.014^\circ$$

Copying and Pasting

You can always copy and paste a previous equation or result into the command window by scrolling up until your desired equation or result is highlighted and simply press **[ENTER]**. This will copy either the equation or result into the command bar to be used to either modify the equation or use the result in another equation.

You can also highlight certain text in the command line and copy/cut and paste within the command line. To highlight text, press and hold the black and white **[F]** key while scrolling to the left or right with the direction keys. The cut/copy/paste options can be found in the **[F1]** menu.